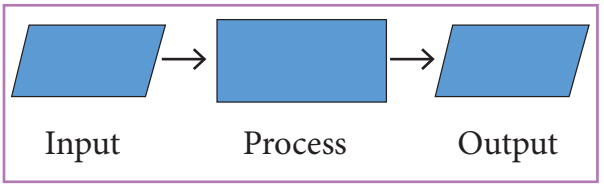
Input	Zero or more quantities to be supplied.
Output	At least one quantity is produced.
Finiteness	Algorithms must terminate after finite number of steps.
Definiteness	All operations should be well defined. For example operations involving division by zero or taking square root for negative number are unacceptable.
Effectiveness	Every instruction must be carried out effectively.
Correctness	The algorithms should be error free.
Simplicity	Easy to implement.
Unambiguous	Algorithm should be clear and unambiguous. Each of its steps and their inputs/outputs should be clear and must lead to only one meaning.
Feasibility	Should be feasible with the available resources.
Portable	An algorithm should be generic, independent of any programming language or an operating system able to handle all range of inputs.
Independent	An algorithm should have step-by-step directions, which should be independent of any programming code.

4.1.2 Writing an Algorithm

Algorithms are generic and not limited to computer alone. It can be used in various real time activities also. Knowingly or unknowingly we perform many algorithms in our daily life such as packing books in school bag, finding shortest path to search a place, scheduling day-to-day activities, preparation for examination, etc. As we know that all programming languages share basic code constructs like conditions and iterations can be used to write an algorithm. A typical algorithm is shown in the following Figure 4.1.



A Typical Algorithm

Example

Consider the example of Coffee preparation. To make coffee, we need to have the following ingredients: Water, milk, coffee powder and sugar. These ingredients are the inputs of an algorithm. Preparing a cup of coffee is called process. The output of this process is coffee.

The procedure for preparing coffee is as follows:

1. Take a bowl with coffee powder
2. Boil the water and pour it into the bowl